



NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA – 769 008

Department	Computer Science & Engineering
Examination	Mid Semester
Program	B.Tech
Session	AUTUMN 2025
Course code & Course Name	CS3007: COMPILER DESIGN
Marks	30
Duration	2:00 Hours

This question paper contains 02 page

- Answer all Questions.
- The marks are indicated against each question. Answer all parts of a question in one place

Q. No.	Questions	Marks	CO*	BL**
1[A]	Define a compiler. Explain the different phases of compilation with a neat diagram. Explain the role of the lexical analyser. What is the purpose of tokenization in compilation?	3	CO1	1
1[B]	Explain predictive parsing. What is the role of FIRST and FOLLOW sets? Construct FIRST and FOLLOW sets for the grammar: $S \rightarrow aABe$ $A \rightarrow Abc \mid b$ $B \rightarrow d$	3	CO2	2
2[A]	Explain operator precedence parsing . Construct an operator precedence table for the grammar of arithmetic expressions. Define the terms precedence relation , associativity , and operator precedence table .	3	CO1	1
2[B]	Differentiate between left recursion and left factoring in grammars. Why is it necessary to eliminate them in parser design? What is the difference between top-down parsing (LL) and bottom-up parsing (LR)? Give one example of each.	3	CO2	2
3[A]	Construct a parse tree and abstract syntax tree (AST) for the input string: $(id + id) * id$ $E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$	3	CO1	1
3[B]	Explain the construction of LR parsing tables. How are conflicts resolved? Define the terms item , viable prefix , and handle in	3	CO3	1

	the context of LR parsing. What are the advantages of LR parsing compared to LL parsing?			
4[A]	Define ambiguity in grammars. Provide an example of an ambiguous grammar. Explain left recursion . How do you eliminate it from a grammar? Explain left factoring with an example.	3	CO2	1
4[B]	What is the role of the symbol table in a compiler? Discuss different symbol table organizations? Explain how scope rules and the block structure of a programming language influence the symbol table organization?	3	CO3	2
5[A]	What is an SLR(1) parser? How does it differ from LR(1) and LALR(1) parsers? Construct the SLR parsing table for the grammar: $S \rightarrow CC$ $C \rightarrow cC \mid d$	3	CO1	1
5[B]	Differentiate between shift-reduce parsing and operator precedence parsing. What is a parsing conflict (shift/reduce and reduce/reduce)? Give an example.	3	CO3	2

CO**: Course Outcome; *BL**: Bloom's Taxonomy Level.